Randy Tang

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SUMMARY

Software engineer with 3+ years building high-throughput, real-time data pipelines processing tens of millions of daily transactions at a global bank. Profitable personal crypto trader (2023–2025) across CEX and DeFi platforms (Kraken, Hyperliquid, Jupiter, Lighter); connected in the crypto trading community. Deep Python proficiency, hands-on experience with event-driven architectures and message queues, and a track record of building trader-facing monitoring and alerting tools.

TECHNICAL SKILLS

Languages: Python, SQL, Java (fundamentals) | **Data/Systems:** High-throughput pipelines, AMPS message queues (pub/sub), concurrency, Pandas, Postgres, SQLite, Docker
Frameworks: Flask, FastAPI (familiar) | **Crypto:** Exchange APIs (Binance, Hyperliquid, Kraken), market data (OI, funding, order flow), DeFi protocols
Practices: Performance optimization, real-time data pipeline design, code review, Git

EXPERIENCE

Software Engineer — Bank of America Aug 2022 – Present

Cirrus OTC Trade Reporting | Python, Pandas, Quartz, AMPS

- Designed and maintained **real-time data pipelines** processing **tens of millions of transactions daily** across multiple services with retries, error handling, and load balancing.
- Built an extensible **reference-data framework**: event-driven pipeline of services connected via **AMPS message queues**, querying multiple external APIs per trade event and handling failures gracefully.
- Implemented a **T+1 trade reconciliation service** validating trade-ID uniqueness across counterparties and issuing corrections at millions of transactions per day.
- Developed a parallelized **high-volume backreporting service**, automating a previously manual process with substantial speed and throughput gains.
- Built a transformer service (internal model → outgoing XML) and response parser to **reconcile transaction statuses** in real time.
- Optimized reliability, latency, and throughput of production services by orders of magnitude; authored Python/Pandas tooling to **analyze 50–100GB datasets**.

Data Analyst (Contract) — Microsoft Mar 2022 – Aug 2022

Viva Topics | Python, Pandas, Jupyter

- Processed and annotated datasets used to train ML models; built Python pipelines to clean, transform, and generate reports on user-collected data for the data science team.

Engineering Intern — Northrop Grumman Fall 2019, Summer 2020

NX, ANSYS

- Designed and released 50+ engineering drawings including flight and test components; performed structural analyses for large assemblies and fastener stacks.

PROJECTS

- **Market Data Dashboard** (Postgres, Docker, Grafana, Twilio) — Ingests **real-time market data** (open interest, funding rates) from **Binance and Hyperliquid APIs**; visualization dashboard with automated SMS alerts on signal triggers.
- **Polymarket Scanner** (Flask, Discord API, SQLite) — Scans Polymarket for **large trades and unusual activity**; real-time alerting pipeline delivering notifications to Discord.
- **Reading Recommendations App** (Flask, Gemini API, SQLite) — LLM-powered daily recommender integrating generative AI APIs over a Project Gutenberg corpus.

EDUCATION

Cornell University, College of Engineering — Ithaca, NY Sep 2017 – Dec 2021

B.S. Computer Science & B.S. Mechanical Engineering | GPA 3.7

- **CUAIR (Cornell Unmanned Aircraft)** — Developed systems for an autonomous plane including obstacle avoidance, object classification, and airdrop; top-performing team at the AUVSI SUAS competition.